Cpts 317. 9/21

So far:

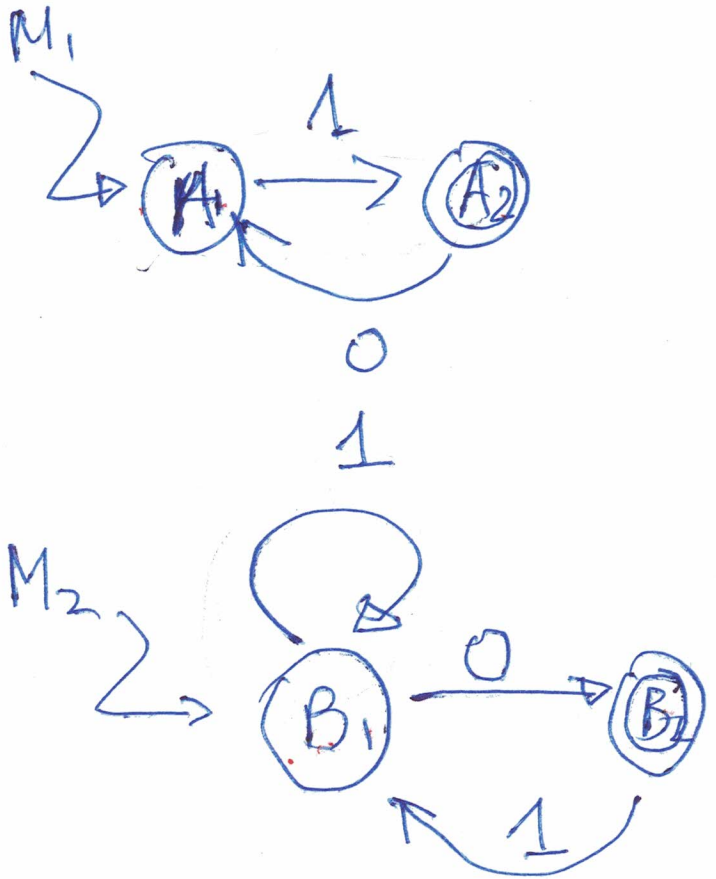
RegExp —— 1, 2, 3, 4.1, 4.2, 4.3.

Structural Induction —— proofs.

DFA —— graph with labels
↑ input symbols.

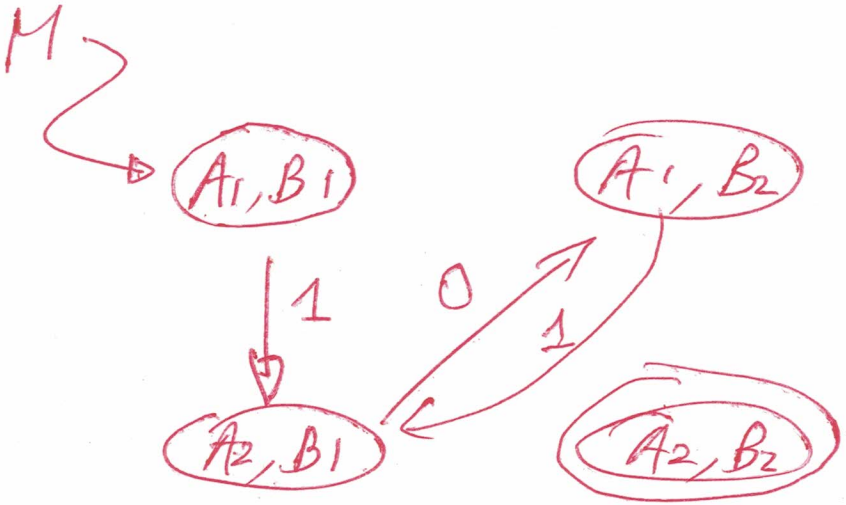
Cartesian Product Construction. (intersection of languages)

Complement Construction (complement of languages).
↑ hw!



Construct $M = M_1 \times M_2$

s.t. $L(M) = L(M_1) \cap L(M_2)$.



$L(M) = \emptyset$.

Assume that: M_1 has n_1 states

M_2 has n_2 states,

then, $M = M_1 \times M_2$ has $n_1 \cdot n_2$ states.

Suppose that M_1 has memory size k_1 bits;

M_2 has " " k_2 ";

then, $M = M_1 \times M_2$ has $k_1 + k_2$ memory size bits.

why? k_1 bits memory $= 2^{k_1}$ states; for M_1

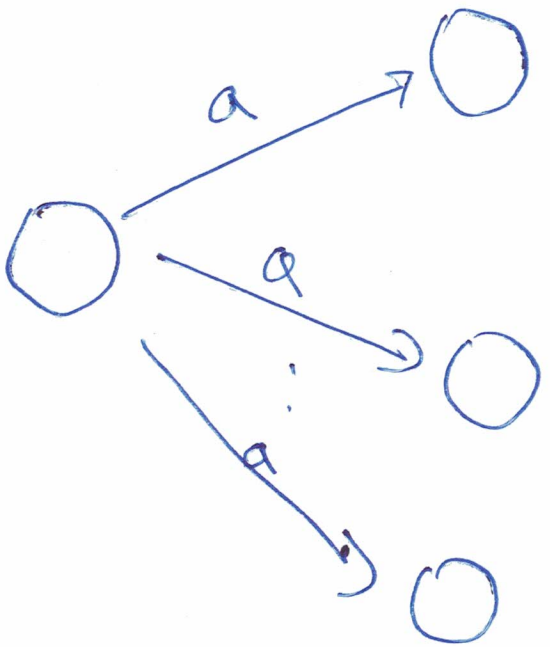
k_2 bits memory $= 2^{k_2}$ states for M_2.

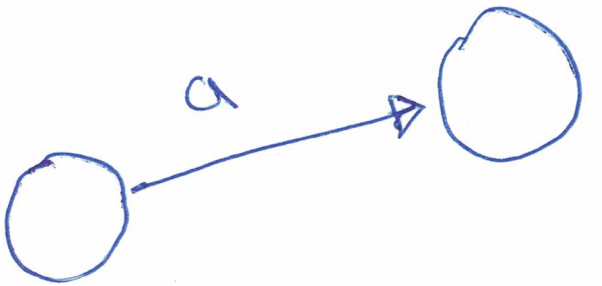
$k_1 + k_2$ bits memory $= 2^{k_1 + k_2} = 2^{k_1} \cdot 2^{k_2}$ states for M.

NFA — nondet. Finite Automata.

NFA vs. DFA

main diff:





Fair Coin X

$$\text{Prob}(X = \text{head}) = 50\%$$

$$\text{Prob}(X = \text{tail}) = 50\%.$$

Pizza with labels

nondet $\neq$ randomness.

theoretical root of nondet. comes from Logic "OR";
theoretical root of randomness comes from probability theory
(measure theory, where probability is defined as the size of a slice of pizza —— no randomness at all.)